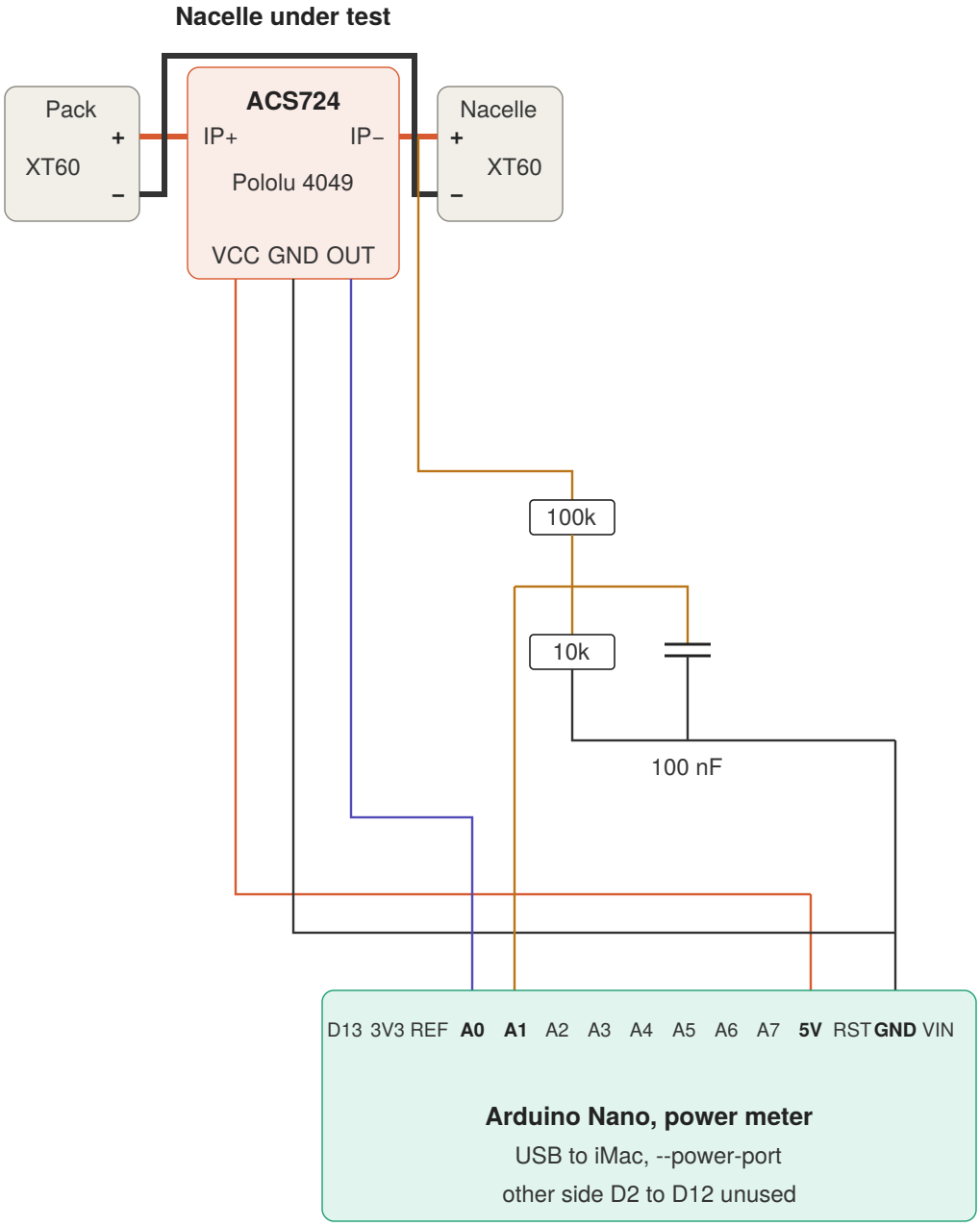
Power Nano, single-channel ACS724 schematic (thrust stand)



Legend

- Pack positive, 12 AWG, through IP+ then IP−, board arrow toward nacelle
- Pack negative, 12 AWG, straight pack to nacelle, never on the sensor board
- VCC, red 22 to 24 AWG, to Nano 5V
- GND, black 22 to 24 AWG, board and divider to Nano GND
- OUT, current signal, to A0 (port channel in live.py)
- Voltage sense: 100k from nacelle-side positive, 10k to GND, junction to A1

100 nF ceramic cap from the junction to GND, on the perfboard at the Nano. FILTER holes stay empty.

Nano top row as printed: D13, 3V3, REF, A0 ... A7, 5V, RST, GND, VIN. A2, A3 and the other row unused.